

CSCE 775: Deep Reinforcement Learning

Project Proposal

The class project gives you the opportunity to use deep reinforcement learning to solve a problem of your choosing. The project will be done in teams of two to three. You are encouraged to incorporate aspects of your own research into your project. While novel research ideas and applications are encouraged, they are not necessary to receive full credit. If you decide not to work on a novel idea, you can replicate results from a reinforcement learning paper of your choosing. However, in addition to replication, you should think of one or two small modifications of the approach and evaluate the results.

Please come to office hours, send me emails, or set up a time to meet with me, individually, to get feedback on your ideas. The use of figures and diagrams in your proposal is encouraged. Use Google Scholar and look at papers from conferences such as the Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), Association for the Advancement of AI (AAAI), Reinforcement Learning Conference (RLC), and International Conference on Machine Learning (ICML) to find papers that are interesting to you.

Only one team member needs to submit the proposal.

Proposal Format

Header (10pts)

CSCE 775: Deep Reinforcement Learning and Search

Project Title

Team Members

Date

Introduction (1-2 paragraphs, 20 pts)

Describe the problem that you are trying to solve and explain why it is interesting.

Approach (1-2 paragraphs, 20 pts)

Describe the methods you are going to use to solve this problem. What is the related work? How does your approach differ from other approaches?

Evaluation (1-2 paragraphs, 20pts)

Describe how you will evaluate your approach. To what baselines will you compare your approach? What metrics are you going to use?

Conclusion (1 paragraph, 20pts)

Summarize the problem you are solving, your approach, and how you plan to evaluate your approach.

References (at least 4 references, 10pts)

Cite relevant papers. You can use Google Scholar to assist with citations.

What to Turn In

Submit a PDF of your project proposal to Blackboard.